Phantom Codes: Hallucination, Accuracy, and Cost in LLM-Based Medical Concept Normalization

Brian K. Fung, PharmD, MPH

Abstract—We evaluate the most recent frontier large language models (LLMs) on medical concept normalization, separating fabrication of non-existent codes from abstention as distinct failure modes, and identify the (model, prompting-mode) configurations that combine high accuracy with low per-prediction cost. We evaluate eight frontier LLMs (Claude Opus 4.7, Sonnet 4.6, Haiku 4.5; GPT-5.5, GPT-4o-mini; Gemini 2.5 Pro, 2.5 Flash, 3 Flash Preview) under three prompting modes (zero-shot, constrained, retrieval-augmented) against string-matching and frozen sentence-transformer baselines. Inputs are Synthea-generated FHIR Conditions under four degradation modes; a six-way outcome taxonomy separates fabrication from abstention. On 125 unique Synthea Conditions across four degradation modes, fabrication ranged from 0 to 5.6% across flagship configurations and from 0 to 12.8% across sub-flagship configurations. Constrained prompting eliminated fabrication for every Anthropic and OpenAI model. Abstention proved more consequential for some models: Gemini 2.5 Pro returned an empty prediction on 88% of zero-shot D4 inputs. Cost-per-correct varied by a factor of 270 across the matrix; GPT-4o-mini in constrained mode achieved the lowest cost at $0.0003 per exact match. Zero-shot LLM prompting is inferior to constrained and retrieval-augmented grounding for medical concept normalization; only the grounded configurations eliminated observed fabrication. Cost-per-correct drives deployment selection: GPT-4o-mini constrained leads at 94.2% top-1 and $0.0003 per correct, with Claude Haiku 4.5 constrained as the higher-accuracy alternative (96.8%, $0.0044). Both dominate the largest frontier model tested (Opus 4.7, $0.0133 per correct).

Index Terms—Natural language processing, International Classification of Diseases, artificial intelligence, medical informatics, reproducibility of results, large language models, hallucination, semantic interoperability.

I. BACKGROUND AND SIGNIFICANCE

Medical concept normalization is the entity-linking task of mapping in-text mentions of biomedical concepts like diagnoses, medications, labs, and procedures to entries in a standardized ontology or controlled vocabulary such as ICD-10-CM, LOINC, or RxNorm [1, 2]. It sits at the foundation of healthcare data interoperability. Errors propagate downstream: a misassigned diagnosis code can shift a hospital's case-mix index, mis-stratify a research cohort, or corrupt the training data of the next generation of models. For most of the last decade, the state of the art for automated coding came from supervised neural classifiers trained on labeled clinical text [3, 4]. The arrival of general-purpose large language models has changed the deployment picture: foundation models can in principle perform code prediction with no task-specific fine-tuning. Two open questions motivated this work. Do the fabrication findings reported on earlier model generations still hold for the frontier models shipping today? And what does the cost side of the deployment tradeoff look like for the configurations that do work?

Concept normalization is also a load-bearing component of semantic interoperability: the property that lets disparate healthcare information systems exchange data in ways that preserve clinical meaning, not just syntactic structure. Modern interoperability stacks (HL7 FHIR, IHE profiles, the OMOP Common Data Model) presume that values in coded fields resolve to entries in controlled vocabularies (ICD-10-CM, LOINC, RxNorm, SNOMED CT); they break silently when those resolutions fail. As clinical data warehouses scale to support secondary use, multi-site cohort construction, and FAIR data practices, automated terminology grounding becomes a critical step in the harmonization pipeline. LLMs are a candidate mechanism for that step. What determines whether they are viable in production is not raw accuracy in isolation but the conjunction of three properties: how often they fabricate codes that do not exist (silently corrupting downstream harmonization), how often they abstain (creating tractable gaps a reviewer can fill), and at what per-prediction cost the resulting pipeline operates at scale.

The literature has documented the fabrication problem but on generations that are now dated. Soroush et al. (2024) reported that GPT-4 achieved only 46% exact match on ICD-9 and 34% on ICD-10, with a substantial fraction of errors being non-existent codes rather than real-but-wrong assignments [5]. Lee and Lindsey documented that LLMs systematically struggle to distinguish real codes from plausibly-formatted fakes [6]. Both findings are an instance of the broader medical-hallucination phenomenon [7, 8], and both predate Anthropic's Opus 4.7, OpenAI's GPT-5.5, and Google's Gemini 3 Flash Preview. Whether the fabrication rates reported on GPT-4 generalize to the current generation has not been re-measured.

Cost is largely absent from the published evaluations. The field's response to hallucination has been architectural: retrieval-grounded pipelines [9, 10], agentic walkers [11], neuro-symbolic verifiers [12], and broader benchmark

B. K. Fung is an independent researcher, Brandon, FL, USA (e-mail: brian@briankfung.com; ORCID: 0000-0002-5951-3770).

expansions [13, 14, 15, 16]. Across this body of work, hallucination is rarely treated as an explicit outcome, and per-prediction cost is rarely reported at all. As LLM-augmented coding moves toward production deployment, the operationally relevant question is which (model, prompting mode) combination achieves a useful accuracy at a tolerable cost. The literature does not yet answer that question.

We present Phantom Codes, whose primary contribution is a reproducible evaluation framework for LLM-based terminology grounding, demonstrated on the most recent frontier-model generation. The framework has four reusable components. *First*, a six-way outcome taxonomy in which fabrication of non-existent codes (hallucination) and abstention (`no_prediction`) are explicit, mutually exclusive buckets; the taxonomy generalizes to any closed-vocabulary grounding task including LOINC and RxNorm. *Second*, a within-model ablation across zero-shot, constrained, and retrieval-augmented prompting modes that holds the model and input fixed and varies only whether a candidate list is provided, isolating "wandering off the menu" from genuine ignorance. *Third*, an abbreviation-stress mode (D4) that replaces canonical display strings with clinical jargon (T2DM, HTN, CKD-3) to expose where string-matching baselines collapse and where genuine semantic mapping is required; D4 functions as a robustness probe for any LLM-grounding pipeline. *Fourth*, per-prediction cost as a first-class outcome alongside accuracy; we report cost-per- correct (USD per exact-match outcome) as the deployment-relevant normalization, computed from per-call token usage against published per-token rates.

The empirical application of this framework spans 24 (model, prompting-mode) configurations across Claude Opus 4.7, Sonnet 4.6, and Haiku 4.5; GPT-5.5 and GPT-4o-mini; and Gemini 2.5 Pro, 2.5 Flash, and 3 Flash Preview — several released within weeks of the evaluation date. The headline matrix runs against Synthea-generated FHIR Bundles [17], a freely-redistributable synthetic patient dataset that gives full reproducibility from a pinned Synthea version and seed.

II. MATERIALS AND METHODS

A. Data

The headline evaluation matrix runs entirely against Synthea-generated FHIR Bundles [17], a freely- redistributable synthetic patient dataset.

B. Cohort and label space

We restrict the cohort to ICD-10-CM codes in the CMS ACCESS Model FHIR Implementation Guide v0.9.6 [18]. The Implementation Guide defines two disease groups: CKM (cardiometabolic, covering diabetes, atherosclerotic cardiovascular disease, and CKD stage 3) and eCKM (extended cardiometabolic, covering hypertension, dyslipidemia, prediabetes, and obesity). The headline matrix is evaluated on a single Synthea-generated cohort of 125 unique FHIR Conditions, each materialized into four degradation modes for a total of 500 EvalRecord items. Cohort generation is fully deterministic: Synthea v4.0.0 [17] is run at a pinned commit SHA with `seed=42`, producing patient bundles whose Conditions are extracted, deduplicated per (patient, ICD code), and filtered to ACCESS-scope codes via bundled value-set definitions. Twelve unique ICD-10-CM codes appear in the cohort: obesity (~24%), prediabetes (~20%), type 2 diabetes variants (~31%), essential hypertension (~14%), and lipid disorders (~10%).

C. Models evaluated

The headline matrix comprises 28 model configurations. These are: three string-matching baselines (exact, fuzzy, TF-IDF), one frozen sentence-transformer retrieval baseline, and 24 frontier LLM configurations across three providers and three prompting modes. Each LLM appears in `zeroshot`, `constrained`, and `rag` configurations.

- Anthropic (3 × 3): Claude Haiku 4.5, Sonnet 4.6, Opus 4.7
- OpenAI (2 × 3): GPT-5.5 (`gpt-5.5-2026-04-23`) and GPT-4o-mini [19]
- Google (3 × 3): Gemini 2.5 Pro, Gemini 2.5 Flash, and Gemini 3 Flash Preview (preview-status, disclosed as such) [20]

Gemini 3.1 Pro Preview was excluded due to insufficient daily quota at our API tier; a partial-coverage analysis appears in Supplementary §S2. Per-token pricing for cost calculations was sourced from each provider's documentation as of the evaluation date [21, 22].

D. Prompting modes

All LLMs are evaluated under three prompting modes that vary in how much external structure is provided.

- **`zeroshot`**: a generic system prompt instructs the model to return the most likely ICD-10-CM code for the input, with no candidate list.
- **`constrained`**: the system prompt additionally includes the full list of CMS ACCESS-scope candidate codes (with display strings) and instructs the model to choose only from that list. Mechanical schema enforcement (Anthropic forced tool use, OpenAI structured outputs, Google `response_schema`) is applied across all modes; in `constrained` mode the menu itself is also passed in-prompt.
- **`rag`**: a frozen sentence-transformer (`all-MiniLM-L6-v2`) retrieves the top-k = 10 ACCESS-scope candidate codes most similar to the per-record input and injects only those into a constrained-style prompt.

The verbatim system and user prompts and per-provider tool-use and schema configurations are reproduced in Supplementary §S1.

E. Outcome taxonomy

Each per-record top-1 prediction is classified into one of six mutually exclusive, exhaustive outcome buckets, ordered from best to worst by deployment relevance:

1. `exact_match`: predicted code equals ground truth.
2. `category_match`: predicted code is in the same ICD-10-CM 3-character category as ground truth (e.g., E11.0 vs. E11.9, both Type 2 diabetes).
3. `chapter_match`: same ICD-10-CM chapter, different category.
4. `out_of_domain`: real ICD-10-CM code with no hierarchical relation to ground truth.
5. `no_prediction`: model returned no usable prediction (empty array, refusal, or transient API failure that exhausted the SDK retry budget). The model fabricated nothing; it abstained.
6. `hallucination`: predicted code does not exist in the FY2026 CMS ICD-10-CM tabular list (mechanically validated against the bundled validator).

The `no_prediction` and `hallucination` buckets capture qualitatively different failure modes and are reported separately. From a downstream-error perspective, abstention is generally preferable to fabrication, since an abstaining model emits no spurious code that downstream systems silently mishandle. Earlier work [5] collapses these into a single error bucket; the six-way split surfaces the distinction that conflation hides.

F. Cost tracking

For every model invocation in our matrix the per-prediction CSV records token usage, latency, and `cost_usd` computed at runtime against the versioned pricing snapshot in `configs/pricing.yaml` [21, 22]. These infrastructure metrics are recorded even when the prediction is wrong and serve as the basis for the cost-per-correct normalization reported in Results. Cost-per-correct (USD per exact-match outcome) combines per-call price and per-call accuracy into a single number; we report it alongside per-call cost because the two diverge when accuracy drops.

G. Statistical analysis

Per-(model, mode) outcome rates are reported as point estimates with 95% Wilson confidence intervals, which we prefer over normal-approximation intervals because both small N and rates near zero or one are common in our matrix (per-cell N = 125, and constrained-mode hallucination rates approach 0%). Within-model paired comparisons across zero-shot, constrained, and RAG modes on identical inputs are reported as exact McNemar tests on discordant prediction pairs. Across-model and across-mode comparisons are reported descriptively with overlapping or non-overlapping CIs. We deliberately do not perform null-hypothesis significance tests across the full grid, because the intended interpretation is comparative deployment relevance rather than detection of any single effect.

III. RESULTS

A. Cohort

The Synthea evaluation cohort comprises 125 unique synthetic FHIR Conditions [17], materialized into all four degradation modes for a total of 500 records. Twelve unique ICD-10-CM codes appear: obesity (E66.9; 24%), prediabetes (R73.03; 20%), type 2 diabetes variants (E11.x; 31% combined), essential hypertension (I10; 14%), and lipid disorders (E78.x; 10%). The matrix evaluates 28 model configurations.

B. Outcome distribution

Two patterns dominate (per-cell N = 125; full table in Supplementary §S2.1). Anthropic constrained and RAG modes achieve zero fabrication across all sizes and all four degradation modes, and GPT-4o-mini and GPT-5.5 in constrained or RAG mode match. Fabrication concentrates instead in zero-shot prompting and varies by model tier. Flagship models (Opus 4.7, GPT-5.5, Gemini 2.5 Pro) range from 0% to 5.6% across all cells, with the maximum from Opus zero-shot D3. Sub-flagship models (Sonnet 4.6, Haiku 4.5, GPT-4o-mini, Gemini Flash variants) range from 0% to 12.8%, with the maximum from GPT-4o-mini zero-shot D2. A pooled median understates the spread, because D1 and D2 inputs expose the source code and constrained or RAG cells are 0% by design. Gemini 2.5 Pro's 0% fabrication rate comes from abstention (95.2% no_prediction at D4 zero-shot), not from correct answering.

C. Failure-mode breakdown: hallucination and abstention

The literature historically conflates two failure modes: fabrication of non-existent codes (hallucination) and abstention (no_prediction). They surface different patterns (Figure 1). Table 1 reports both rates per cell as hallucination % / no_prediction %. Per-cell N = 125; Wilson 95% CIs are ±3pp at 0%, ±5pp at 5%, and ±8 to ±9pp at 50% (full CIs in §S2.2 to §S2.3).

Within-model paired comparison (zero-shot versus constrained, McNemar tests on discordant pairs) confirms the constrained-mode reduction is statistically significant for every Anthropic and OpenAI model with nonzero zero-shot hallucination. Gemini 2.5 Pro zero-shot exhibits a near-complete abstention pattern (95.2% no_prediction at D4) with 0% fabrication. The failure mode is abstention, not fabrication, and constrained and RAG cut but do not eliminate it. Whether this reflects model behavior or a wrapper-level interaction with the reasoning-token budget is unresolved (see §Discussion limitations). Gemini 2.5 Flash and Gemini 3 Flash Preview zero-shot abstain rather than fabricate. When these models do return a code under D2 to D4 stress, it is almost always real (hallucination $\leq$0.8% across all cells); the dominant failure is empty-prediction, which rises to 39 to 42% under D4 abbreviation stress. Full 24-LLM tables in Supplementary §S2.2 to §S2.3.

TABLE I
FAILURE-MODE BREAKDOWN BY MODEL AND DEGRADATION MODE. EACH CELL REPORTS HALLUCINATION % / NO_PREDICTION %.

Model	Mode	D1_full	D2_no_code	D3_text_only	D4_abbreviated
claude-haiku-4-5	constrained	0.0 / 0.0	0.0 / 0.0	0.0 / 0.0	0.0 / 0.0
claude-haiku-4-5	zeroshot	0.0 / 0.0	3.2 / 0.0	4.8 / 0.0	3.2 / 0.0
claude-sonnet-4-6	constrained	0.0 / 0.0	0.0 / 0.0	0.0 / 0.0	0.0 / 0.0
claude-sonnet-4-6	zeroshot	0.0 / 0.0	0.8 / 0.0	6.4 / 0.0	3.2 / 0.0
claude-opus-4-7	constrained	0.0 / 0.0	0.0 / 0.0	0.8 / 0.8	0.0 / 0.0
claude-opus-4-7	zeroshot	0.8 / 0.0	0.0 / 0.0	5.6 / 0.0	1.6 / 1.6
gpt-5.5	constrained	0.0 / 0.0	0.0 / 0.0	0.0 / 0.0	0.0 / 0.0
gpt-5.5	zeroshot	0.8 / 0.8	0.0 / 0.0	0.8 / 0.8	0.0 / 0.0
gpt-4o-mini	constrained	0.0 / 0.0	0.0 / 0.0	0.0 / 0.0	0.0 / 0.0
gpt-4o-mini	zeroshot	0.8 / 0.0	12.8 / 0.0	9.6 / 0.0	7.2 / 0.0
gemini-2.5-pro	zeroshot	0.0 / 77.6	0.0 / 85.6	0.0 / 94.4	0.0 / 95.2
gemini-2.5-pro	constrained	0.0 / 54.4	0.0 / 32.0	0.0 / 67.2	0.0 / 60.8
gemini-2.5-flash	zeroshot	0.0 / 0.0	0.0 / 11.2	0.8 / 37.6	0.8 / 42.4
gemini-3-flash-preview	zeroshot	0.0 / 22.4	0.0 / 28.0	0.0 / 41.6	0.0 / 39.2

D. Top-1 vs top-5 exact-match lift

Many configurations have the right answer in their top-5 candidates even when their top-1 pick is wrong. Table 2 reports top-1 accuracy, top-5 accuracy, and the resulting lift across selected configurations.

Two patterns are visible. LLMs in *constrained* mode saturate at top-1 (lift $\leq$6pp), so there is little additional value in human-in-the-loop top-5 review for those configurations. Non-LLM baselines and sub-frontier LLMs in zero-shot mode show much larger lifts (20 to 38pp), suggesting that workflows surfacing top-5 candidates to a human reviewer would substantially improve net accuracy. Gemini 2.5 Pro's zero lift is consistent with the abstention-dominated failure mode, since there is no candidate to be lifted into top-5 when the model returns nothing.

E. Cost per correct prediction

Most prior LLM medical-coding benchmarks report accuracy in isolation [5, 13, 14]. The deployment decision, however, hinges on the joint distribution of API cost per call, per-call accuracy, and the rate of failure modes that drive downstream QA cost. Table 3 reports per-call API cost across selected (model, prompting-mode) configurations from the headline matrix; the full 28-configuration breakdown, per-bucket cost decomposition, break-even analysis (LLM and QA versus a human coder), and annual cost projections at deployment scale are reported in Supplementary §S5.

† One of 500 attempted gpt-5.5 zero-shot calls exhausted the SDK retry budget on a transient API error and is excluded from the cost total. We report n=499 as-is rather than re-running, since transient API failures are an inherent property of production LLM deployment and re-running would obscure that signal.

‡ Gemini 2.5 Pro zero-shot completed 424 of 500 attempted calls under our wrapper's `max_output_tokens=1024` setting. The remaining 76 returned empty responses consistent with reasoning-token budget exhaustion (counted as `no_prediction` in §4.3, not as cost-bearing calls). The 424 figure reflects calls that produced billable output; cost normalization uses that denominator.

A 270× spread separates the cheapest configuration (Gemini 2.5 Flash zeroshot at $0.00005/call) from the most expensive (Claude Opus 4.7 zeroshot at $0.01354/call). Per-call price varies more dramatically across the matrix than per-call accuracy, so cost-per-correct prediction is dominated by per-call price for any configuration that achieves $\geq$80% top-1 accuracy. One caveat: the cost numbers reported here are *automated-coding cost*, not the broader role of clinical informaticists. Any break-even analysis is *deployment cost guidance*, not an *automation feasibility verdict*. Error tolerance is workflow-specific, and a hallucinated code that flows into claims billing has different downstream cost than the same hallucination in a research cohort definition. The full caveats are in Supplementary §S5.

Cost-per-correct ($ per exact-match outcome) collapses per-call price and accuracy into one deployment-ready number; the Top-1 and Halluc columns surface all three deployment dimensions in one row. Plotted as a Pareto frontier in Figure 2 and tabulated in Table 4 below. Sorted by $/correct ascending; n=500 per row.

Reading the table top-down, the cheaper rows are all 4 to 19pp lower on top-1 or carry nonzero hallucination. GPT-4o-mini constrained leads on deployment-relevance: 94.2% top-1, 0% hallucination, $0.0003 per correct, the cheapest configuration that achieves $\geq$94% top-1. Claude Haiku 4.5 constrained adds 2.6pp accuracy at roughly 14× the cost. Claude Opus 4.7 constrained's 0.6pp accuracy edge does not justify its 44× cost: the largest frontier model is not the deployment leader. Gemini 2.5 Pro's elevated $/correct reflects abstention behavior, not a normal cost-accuracy tradeoff.

F. Outcome distribution under D4 abbreviation stress

D4_abbreviated is the strongest stress test in the matrix. Explicit ICD codes and canonical display strings are stripped, and clinical entities are replaced with jargon ("T2DM", "HTN", "CKD-3"). String-matching baselines collapse (fuzzy and TF-IDF both drop ~30pp from D3

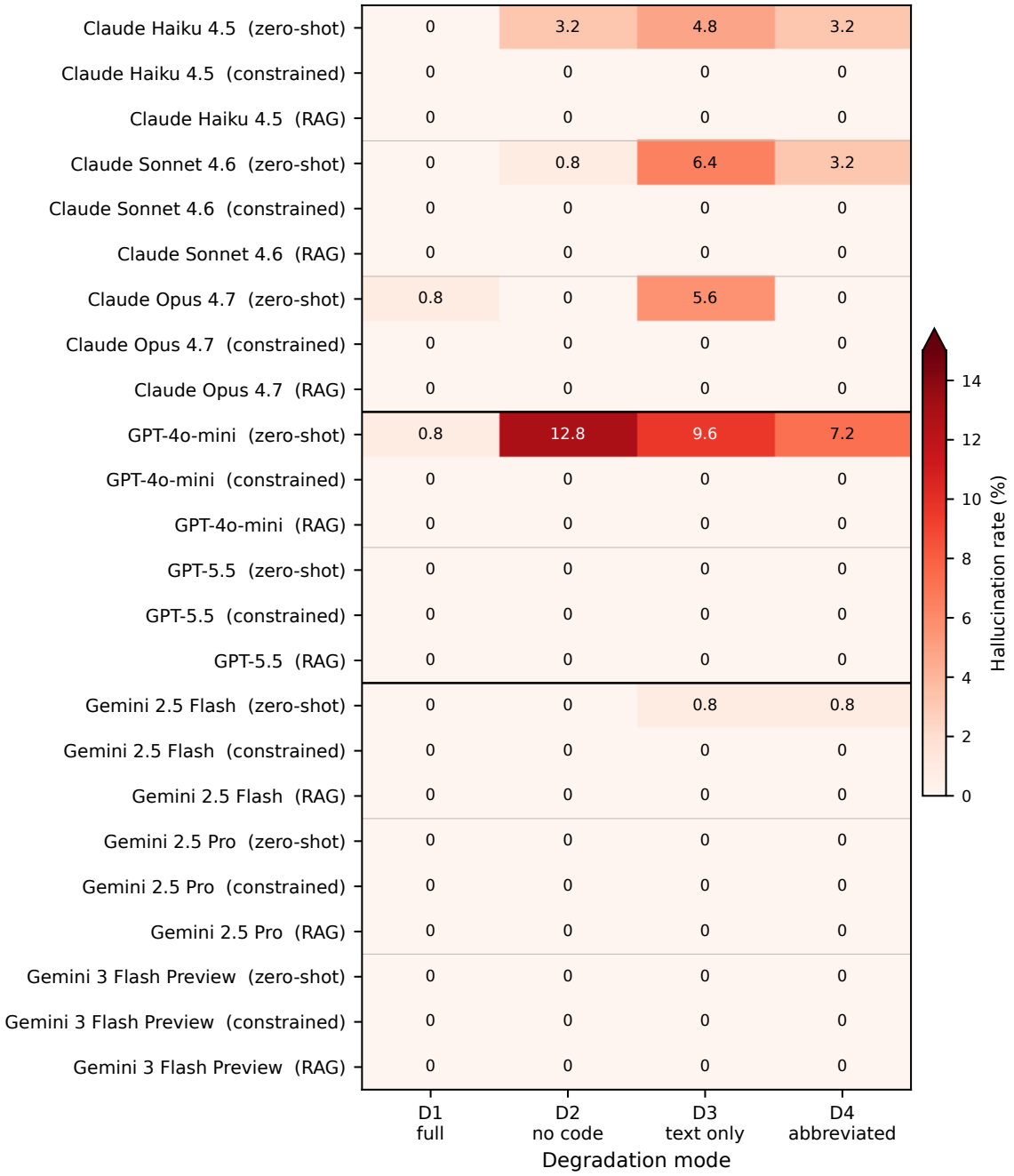


Fig. 1. Hallucination rate (%) by model, prompting mode, and degradation mode. Each row is one (LLM, prompting mode) configuration; columns are the four degradation modes. Cell color encodes the percentage of D-mode predictions that were fabricated codes (predictions that do not exist in the FY2026 CMS-published ICD-10-CM tabular list). Constrained and RAG rows are uniformly zero across all Anthropic and OpenAI models. Zero-shot fabrication concentrates in sub-flagship models, peaking at 12.8% for GPT-4o-mini under D2.

to D4), so remaining top-1 accuracy must come from semantic mapping. Anthropic constrained mode holds up under D4 (Haiku 93.6%, Sonnet 90.4%, Opus 86.4% top-1, versus 100% at D1), and the failure mode is exclusively `category_match`. Frontier zero-shot LLMs show a 5 to 15pp top-1 drop from D1 to D4 but maintain low hallucination. Gemini 2.5 Pro abstention worsens monotonically from D1 to D4 across all three prompting modes. Figure 3 shows the full six-bucket outcome distribution per configuration under D4 stress.

IV. DISCUSSION

A. Principal findings

The headline result is that fabrication is bounded in modern frontier LLMs and stratifies cleanly by model tier and prompting mode, while abstention is the more consequential failure mode for some models. The 6-way taxonomy is what makes the second pattern visible. Across

TABLE II
TOP-1 VS TOP-5 EXACT-MATCH LIFT ACROSS SELECTED CONFIGURATIONS.

Model	Top-1	Top-5	Lift
claude-haiku-4-5 (constrained)	96.8%	100.0%	+3.2pp
claude-sonnet-4-6 (constrained)	94.6%	100.0%	+5.4pp
claude-opus-4-7 (constrained)	94.8%	99.8%	+5.0pp
gpt-4o-mini (constrained)	94.2%	100.0%	+5.8pp
gpt-5.5 (constrained)	93.6%	100.0%	+6.4pp
sentence-transformer:retrieval	69.0%	90.6%	+21.6pp
baseline:tfidf	45.2%	83.2%	+38.0pp
claude-sonnet-4-6 (zeroshot)	75.4%	95.8%	+20.4pp
gpt-5.5 (zeroshot)	88.8%	98.8%	+10.0pp
gemini-2.5-pro (zeroshot)	11.8%	11.8%	+0.0pp

TABLE III
PER-CALL API COST ACROSS SELECTED (MODEL, PROMPTING-MODE) CONFIGURATIONS FROM THE HEADLINE MATRIX. N = NUMBER OF COMPLETED CALLS (OUT OF 500 ATTEMPTED). †, ‡ EXPLAINED IN THE FOOTNOTES BELOW.

Model × Mode	n	$ total	$ per call (mean)
claude-opus-4-7:zeroshot	500	6.77	0.01354
claude-sonnet-4-6:zeroshot	500	3.41	0.00682
claude-haiku-4-5:constrained	500	2.14	0.00428
gpt-5.5:zeroshot †	499	3.59	0.00720
gpt-4o-mini:constrained	500	0.14	0.00028
gemini-2.5-pro:zeroshot ‡	424	0.41	0.00096
gemini-2.5-flash:zeroshot	500	0.03	0.00005

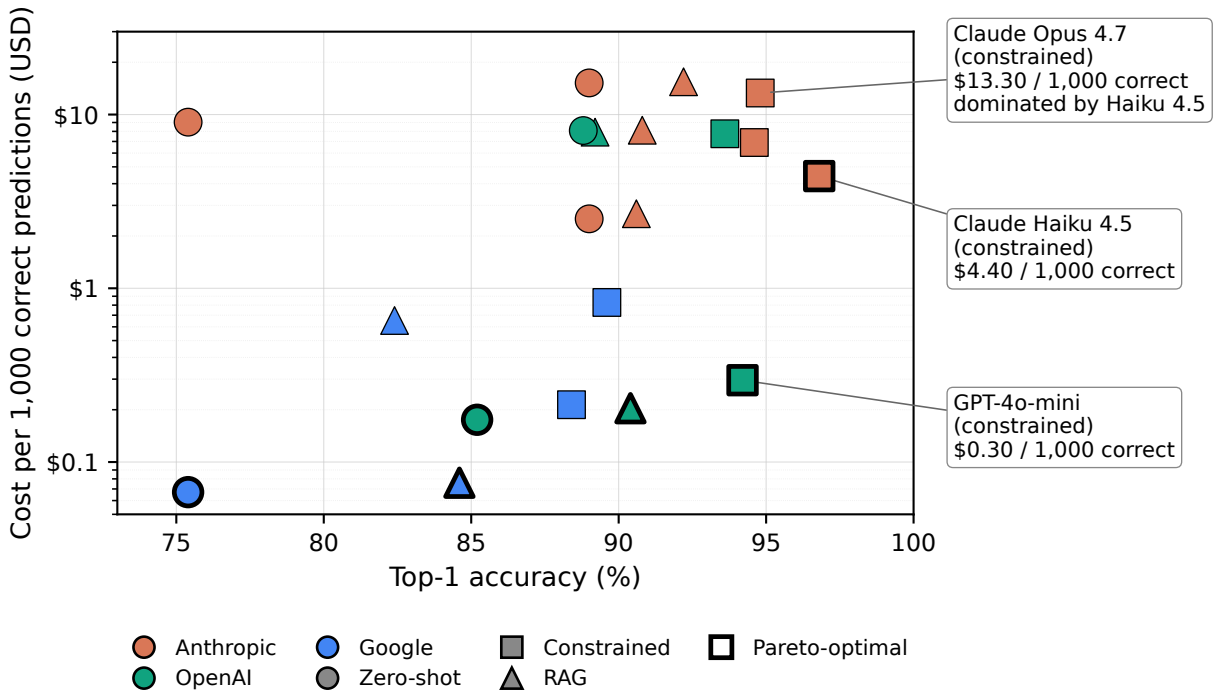


Fig. 2. Cost per 1,000 correct predictions (USD, log scale) versus top-1 accuracy for every (LLM, prompting mode) configuration achieving ≥75% top-1 accuracy. Color encodes provider; marker shape encodes prompting mode. Pareto-optimal configurations are drawn with a heavy black border. GPT-4o-mini constrained and Claude Haiku 4.5 constrained sit on the deployment frontier; Claude Opus 4.7 constrained is dominated despite competitive accuracy because its per-correct cost is 30 to 40 times higher.

the 24 LLM configurations, hallucination ranges from 0% to 5.6% for flagship models (Opus 4.7, GPT-5.5, Gemini 2.5 Pro) and from 0% to 12.8% for sub-flagship models, with the upper bound from GPT-4o-mini zero-shot D2. Hallucination behaves as a property of the prompting setup rather than a fixed property of the model. The shift from zero-shot to constrained prompting eliminates fabrication entirely for every Anthropic and OpenAI model that shows any nonzero zero-shot hallucination, and within-model paired McNemar tests confirm significance at n=125.

TABLE IV
COST PER CORRECT PREDICTION. *CONSTR.* = CONSTRAINED. *GEMINI 2.5 PRO CONSTRAINED REFLECTS ABSTENTION PATTERN, NOT ERROR RATE; SEE §4.3 (FAILURE-MODE BREAKDOWN).

Model	Mode	Top-1	Halluc	Total	$/correct
Gemini 2.5 Flash	rag	84.6%	0.0%	$0.03	$0.0001
GPT-4o-mini	zeroshot	85.2%	7.6%	$0.07	$0.0002
GPT-4o-mini	rag	90.4%	0.0%	$0.09	$0.0002
Gemini 2.5 Flash	constr.	88.4%	0.0%	$0.09	$0.0002
GPT-4o-mini	constr.	94.2%	0.0%	$0.14	$0.0003
Gemini 3 Flash Preview	constr.	89.6%	0.0%	$0.37	$0.0008
Claude Haiku 4.5	rag	90.6%	0.0%	$1.22	$0.0027
Claude Haiku 4.5	constr.	96.8%	0.0%	$2.14	$0.0044
Claude Sonnet 4.6	constr.	94.6%	0.0%	$3.26	$0.0069
GPT-5.5	constr.	93.6%	0.0%	$3.63	$0.0078
Gemini 2.5 Pro*	constr.	46.2%	0.0%	$1.57	$0.0068
Claude Opus 4.7	constr.	94.8%	0.0%	$6.31	$0.0133

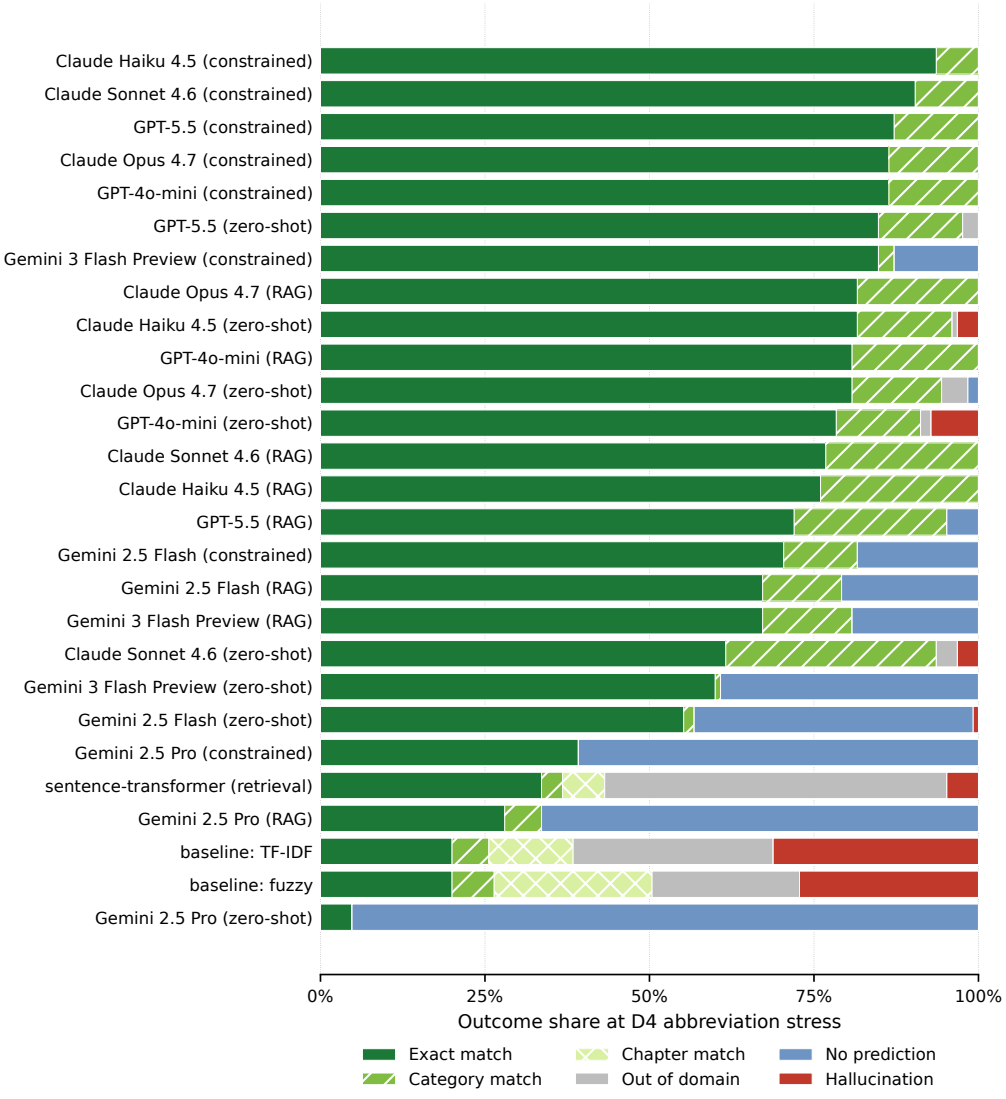


Fig. 3. Outcome distribution per (model, prompting mode) configuration under D4 abbreviation stress. Each horizontal bar sums to 100% across the six outcome buckets defined in §2.5; bars are ordered by exact_match share descending. Anthropic and OpenAI constrained configurations cluster at the top, dominated by exact_match with a small category_match remainder. Gemini 2.5 Pro at the bottom is dominated by no_prediction (abstention), distinct from the hallucination-heavy string-matching baselines (`baseline:fuzzy`, `baseline:tfidf`) below it.

The string-matching baselines and the frozen sentence-transformer baseline (45 to 69% top-1) sit well below every constrained-mode LLM on the fabrication-versus-accuracy frontier, confirming that constrained frontier LLMs offer real gains over lexical and embedding-only approaches at this cohort scale. Cost-per-correct collapses the deployment choice to a small frontier: GPT-4o-mini constrained at \$0.0003 and Claude Haiku 4.5 constrained at \$0.0044 dominate every more expensive configuration tested.

B. Comparison with prior work

The closest direct precedent is Soroush et al. (2024) [5], who benchmarked four 2023–2024 vintage LLMs on direct ICD code querying. Our matrix differs from theirs in three ways relevant to deployment. The first is model generation. Their headline numbers are GPT-4 (46% exact match on ICD-9, 34% on ICD-10); our matrix includes Opus 4.7, GPT-5.5, and Gemini 3 Flash Preview on the same task type. Every Anthropic and OpenAI configuration we evaluated exceeds 80% top-1 even under the D4 abbreviation stress, which we attribute to cohort-scope restriction, generational LLM improvements, and structured-output prompting. The second is per-prediction cost. Soroush et al. and the broader hallucination-mitigation literature [9, 10, 11, 12] do not report per-prediction cost; we do, and cost-per-correct collapses the choice to a small Pareto frontier (GPT-4o-mini constrained at \$0.0003 and Claude Haiku 4.5 constrained at \$0.0044). The third is the outcome taxonomy: hallucination is an explicit, mutually-exclusive bucket, abstention is split out via the 6-way taxonomy, and the within-model zero-shot vs. constrained vs. RAG contrast is reported as a controlled ablation that separates "wandering off the menu" from genuine ignorance. Neuro-symbolic alternatives [11, 12] report hallucination control through architectural constraints; our 0% Anthropic constrained-mode rate sits below their reported neural-baseline band, suggesting that modern frontier LLMs under structured-output constraints have closed much of the fabrication gap those systems were designed to fix. Full comparison is in Supplementary §S2.

C. Implications for clinical deployment

The 6-way taxonomy reframes the deployment question. Top-1 accuracy framing treats every wrong answer equally, but the cost of a hallucinated code (downstream systems silently mishandle it, propagating into billing, research, and quality reporting) is qualitatively different from a near-miss within the correct ICD chapter (corrected by a reviewer in seconds), which is in turn different from an abstention (re-coding from scratch, but emitting no spurious artifact).

The deployment pattern supported by these results is therefore LLM-augmented coding with terminologist or clinical-informaticist review, not autonomous LLM coding. Two findings concretely support this. For constrained-mode Anthropic and OpenAI configurations (0% hallucination, ~5% category-match, ~0% abstention), the QA fraction is dominated by fast category-match adjudication, a clinically straightforward task an experienced coder can resolve in seconds, which beats human-only coding across all sensitivity-sweep parameter ranges considered (Supplementary §S5). The top-1 vs top-5 lift table (§4.4) also shows that trained-classifier and sub-frontier zero-shot LLM workflows (top-1 in the 45 to 75% range) benefit substantially from surfacing top-5 candidates to a reviewer, lifting net accuracy 20 to 38 percentage points. Constrained-mode frontier LLMs already saturate top-1 (lift $\leq$6pp) and shift the reviewer's role from candidate selection to category-match adjudication. Either architecture keeps a clinical adjudicator in the decision loop where accuracy matters most.

D. Implications for semantic interoperability

Automated terminology grounding is a load-bearing component of modern interoperability pipelines: HL7 FHIR, IHE profiles, and the OMOP Common Data Model all presume that coded fields resolve to entries in controlled vocabularies, and they degrade silently when those resolutions fail. The 6-way outcome taxonomy has three practical consequences for pipeline design. *First*, the `hallucination` bucket is a data-quality red flag for any harmonization step: a fabricated code propagates into the harmonized dataset undetected, silently corrupting cohort definitions, billing aggregates, and downstream training data. The `no_prediction` bucket, by contrast, is a tractable gap that surfaces as a missing value a reviewer or fallback heuristic can resolve. Treating these as the same failure mode (as some prior work does) under-counts the silent-corruption risk. *Second*, cost-per-correct is a budget input for interoperability-pipeline design. A pipeline processing 250,000 conditions/year at \$0.0003 per correct (GPT-4o-mini constrained) costs \$70/year in API spend; the same pipeline at \$0.0133 per correct (Opus 4.7 constrained) costs \$3,300/year for the same accuracy floor. The choice is not "which model is best" but "which (model, prompting mode) point on the Pareto frontier matches the deployment constraints." *Third*, the 6-way taxonomy generalizes as a reusable yardstick for any vocabulary-grounding component, not just ICD-10-CM: LOINC for labs, RxNorm for medications, SNOMED CT for broader clinical concepts. The bundled FY2026 CMS validator that distinguishes hallucination from out-of-domain in our matrix has direct analogs in the other vocabulary registries.

E. Limitations

The v1 headline matrix evaluates n=125 unique Synthea Conditions. Per-cell Wilson 95% CIs on a 10% rate are correspondingly ±5pp, and v2 replication at n=2000 will tighten intervals by approximately 2×. Gemini 2.5 Pro showed an 88% no_prediction rate on zero-shot D4 with most empty rows showing zero output tokens, consistent with reasoning-token budget exhaustion under our wrapper's `max_output_tokens=1024` setting. Whether this is a fixable wrapper interaction or a genuine model limitation

is the subject of a follow-up investigation blocked by Tier-1 daily quota at v1 submission. Gemini 3.1 Pro Preview was excluded for the same quota reason. The benchmark covers English ICD-10-CM only; extension to LOINC and RxNorm via the same taxonomy is planned for v2.

F. Future directions

Two extensions follow directly. The first is v2 vocabulary expansion to LOINC and RxNorm; the 6-way taxonomy and eval runner already support arbitrary FHIR resource types. The second is rationale evaluation following Li et al. [15], in which each LLM is asked for a per-prediction rationale to support faithfulness and plausibility evaluation alongside the outcome-bucket framework. A v2 cohort with broader vocabulary and more realistic clinical text (e.g., discharge summaries, ED notes) is also needed to establish the true effect size of fabrication risk on real clinical inputs.

V. CONCLUSION

Across a 24-LLM-configuration evaluation matrix on n=125 Synthea-generated FHIR Conditions, fabrication of non-existent ICD-10-CM codes ranged from 0% to 5.6% for flagship models (Opus 4.7, GPT-5.5, Gemini 2.5 Pro) and from 0% to 12.8% for sub-flagship models. Constrained prompting eliminated fabrication entirely for every Anthropic and OpenAI model. Hallucination behaves as a controllable property of the prompting setup at this scale rather than a fixed model limitation. Abstention, separated from fabrication via the 6-way taxonomy, proved the more consequential failure mode for some models.

Two caveats temper the headline. The n=125 cohort and the narrow 12-code ACCESS-Model vocabulary limit per-cell precision and bound the population that surfaces fabrication. The D4 abbreviation-substitution stress is only one proxy for real-world hallucination triggers; multi-condition narratives, ambiguous documentation, code-specificity ambiguity, and long-tail rare diagnoses are not exercised here. A v2 cohort with broader vocabulary and more realistic clinical text (e.g., discharge summaries, ED notes) is needed to establish the true effect size of fabrication risk on real clinical inputs.

Two recommendations follow from the matrix. Zero-shot LLM prompting is inferior to constrained and retrieval-augmented grounding for medical concept normalization, since every observed fabrication occurred under zero-shot, and only the grounded configurations eliminated it. Among grounded configurations, cost-per-correct rather than accuracy alone should drive selection, because clinical coding errors are downstream-costly and the cheaper model can win on the combined metric. The deployment frontier consists of GPT-4o-mini constrained (94.2% top-1, $0.0003 per correct, the cheapest configuration at $\geq$94% accuracy) and Claude Haiku 4.5 constrained (96.8%, $0.0044 per correct, an additional 2.6pp accuracy at roughly 14× the cost). Both dominate the largest frontier model (Opus 4.7 constrained, 94.8%, $0.0133). The supported workflow is LLM-augmented coding with terminologist review.

ACKNOWLEDGMENTS

Use of AI tools. This work used Anthropic's Claude (Claude Code, model Claude Opus 4.7) as a coding, drafting, and figure-design assistant during the development of the benchmark codebase, the manuscript prose, and the figures. The author conceived and designed the study, defined the experimental framework and outcome taxonomy, conducted all analyses, and reviewed and verified the AI-generated content. The author retains full responsibility for the content, methodology, results, and any errors.

COMPETING INTERESTS

The author declares no competing interests.

FUNDING

This work received no external funding. LLM API access (Anthropic, OpenAI, Google) for the evaluation matrix was self-funded by the corresponding author; per-call costs are itemized in the Results section.

DATA AVAILABILITY

The Synthea-generated FHIR Bundles used for the headline evaluation are freely redistributable and will be released as `benchmarks/synthetic_v1/` alongside the paper at https://github.com/briankfung/phantom-codes. All source code, including the data-preparation pipeline, degradation logic, ICD-10-CM validator, eval runner, prompts, and baselines, is publicly available at the same repository under the MIT License.

CODE AVAILABILITY

All source code, configuration files, synthetic test fixtures, and the manuscript source are publicly available at https://github.com/briankfung/phantom-codes under the MIT License. The repository includes the full evaluation pipeline and the Synthea generation configuration needed to reproduce the headline matrix end-to-end.

REFERENCES

[1] Z. Miftahutdinov, A. Kadurin, R. Kudrin, and E. Tutubalina, "Medical concept normalization in clinical trials with drug and disease representation learning," *Bioinformatics*, vol. 37, no. 21, pp. 3856–3864, 2021. DOI: 10.1093/bioinformatics/btab474 [Online]. Available: https://doi.org/10.1093/bioinformatics/btab474

[2] D. Xu and T. Miller, "A simple neural vector space model for medical concept normalization using concept embeddings," *J Biomed Inform*, vol. 130, p. 104 080, 2022. DOI: 10.1016/j.jbi.2022.104080 [Online]. Available: https://doi.org/10.1016/j.jbi.2022.104080

[3] J. Mullenbach, S. Wiegreffe, J. Duke, J. Sun, and J. Eisenstein, "Explainable prediction of medical codes from clinical text," in *Proceedings of NAACL-HLT*, 2018. [Online]. Available: https://aclanthology.org/N18-1100/

[4] C.-W. Huang, S.-C. Tsai, and Y.-N. Chen, "PLM-ICD: Automatic ICD coding with pretrained language models," in *Proceedings of the 4th Clinical NLP Workshop (NAACL)*, 2022. [Online]. Available: https://aclanthology.org/2022.clinicalnlp-1.2/

[5] A. Soroush, B. S. Glicksberg, E. Zimlichman, Y. Barash, R. Freeman, A. W. Charney, G. N. Nadkarni, and E. Klang, "Large language models are poor medical coders — benchmarking of medical code querying," *NEJM AI*, 2024. DOI: 10.1056/AIdbp2300040 [Online]. Available: https://ai.nejm.org/doi/full/10.1056/AIdbp2300040

[6] S. A. Lee and T. Lindsey, *Can large language models abstract medical coded language?* 2024. arXiv: 2403.10822. [Online]. Available: https://arxiv.org/abs/2403.10822

[7] Y. Kim, H. Jeong, S. Chen, et al., *Medical hallucinations in foundation models and their impact on healthcare*, 2025. arXiv: 2503.05777. [Online]. Available: https://arxiv.org/abs/2503.05777

[8] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. Bang, A. Madotto, and P. Fung, "Survey of hallucination in natural language generation," *ACM Comput Surv*, vol. 55, no. 12, 2023. DOI: 10.1145/3571730 [Online]. Available: https://dl.acm.org/doi/10.1145/3571730

[9] K. Das Baksi, E. Soba, J. J. Higgins, R. Saini, J. Wood, J. Cook, J. Scott, N. Pudota, T. Weninger, E. Bowen, and S. Bhattacharya, *MedCodER: A generative AI assistant for medical coding*, 2024. arXiv: 2409.15368. [Online]. Available: https://arxiv.org/abs/2409.15368

[10] P. Sarvari, Z. Al-fagih, A. Abou-Chedid, P. Jewell, R. Taylor, and A. Imtiaz, "Challenges and solutions in applying large language models to guideline-based management planning and automated medical coding in health care: Algorithm development and validation," *JMIR Biomed Eng*, vol. 10, e66691, 2025. DOI: 10.2196/66691 [Online]. Available: https://biomedeng.jmir.org/2025/1/e66691

[11] A. Motzfeldt, J. Edin, C. L. Christensen, C. Hardmeier, L. Maaløe, and A. Rogers, "Code like humans: A multi-agent solution for medical coding," in *Findings of EMNLP*, 2025. arXiv: 2509.05378. [Online]. Available: https://arxiv.org/abs/2509.05378

[12] Y. Yu et al., *Hybrid-Code v2: Zero-hallucination clinical ICD-10 coding via neuro-symbolic verification and automated knowledge base expansion*, 2025. arXiv: 2512.23743. [Online]. Available: https://arxiv.org/html/2512.23743

[13] V. A. Almeida, V. de Camargo, R. Gómez-Bravo, E. van der Haring, K. van Boven, M. Finger, and L. F. Lopez, "Large language models as medical code selectors: A benchmark using the International Classification of Primary Care," in *NeurIPS Workshop on GenAI for Health*, 2025. arXiv: 2507.14681. [Online]. Available: https://arxiv.org/abs/2507.14681

[14] U. Singh, S. Deshwal, N. Dube, and A. Sharma, *MAX-EVAL-11: A comprehensive benchmark for evaluating large language models on full-spectrum ICD-11 medical coding*, medRxiv, 2025. DOI: 10.1101/2025.10.30.25339130 [Online]. Available: https://www.medrxiv.org/content/10.1101/2025.10.30.25339130v1

[15] M. Li, V. Schlegel, T. Mu, W. Oyewusi, K. Kang, and G. Nenadic, *Evaluation and LLM-Guided learning of ICD coding rationales*, 2025. arXiv: 2508.16777. [Online]. Available: https://arxiv.org/abs/2508.16777

[16] A. Gershon, S. Soffer, G. N. Nadkarni, and E. Klang, *Automatic ICD coding using LLMs: A systematic review*, medRxiv, 2025. DOI: 10.1101/2025.07.30.25330916 [Online]. Available: https://www.medrxiv.org/content/10.1101/2025.07.30.25330916v1

[17] J. Walonoski, M. Kramer, J. Nichols, A. Quina, C. Moesel, D. Hall, C. Duffett, K. Dube, T. Gallagher, and S. McLachlan, "Synthea: An approach, method, and software mechanism for generating synthetic patients and the synthetic electronic health care record," *J Am Med Inform Assoc*, vol. 25, no. 3, pp. 230–238, 2018. DOI: 10.1093/jamia/ocx079 [Online]. Available: https://academic.oup.com/jamia/article/25/3/230/4098271

[18] Centers for Medicare and Medicaid Services, *ACCESS Model FHIR Implementation Guide v0.9.6*, FHIR Implementation Guide, 2026. [Online]. Available: https://dsacms.github.io/cmmi-access-model/

[19] OpenAI, *Latest model guide*, OpenAI Developer Documentation, Accessed 2026-05-03, 2026. [Online]. Available: https://developers.openai.com/api/docs/guides/latest-model

[20] Google, *Gemini 3 model documentation*, Google AI for Developers Documentation, Accessed 2026-05-03, 2026. [Online]. Available: https://ai.google.dev/gemini-api/docs/gemini-3

[21] OpenAI, *Pricing*, OpenAI Developer Documentation, Accessed 2026-05-03, 2026. [Online]. Available: https://developers.openai.com/api/docs/pricing

[22] Google, *Gemini API pricing*, Google AI for Developers Documentation, Accessed 2026-05-03, 2026. [Online]. Available: https://ai.google.dev/gemini-api/docs/pricing

Brian K. Fung (PharmD, MPH) is an independent researcher based in Brandon, Florida, USA. He holds a Doctor of Pharmacy (PharmD) and a Master of Public Health (MPH), and his research focuses on the safe and cost-effective deployment of large language models in healthcare, with current emphasis on medical concept normalization, semantic interoperability, and reproducible evaluation infrastructure for clinical NLP. He declares no competing interests.